

CHAPTER C29

WIND LOADS ON BUILDING APPURTENANCES AND OTHER STRUCTURES: MAIN WIND FORCE RESISTING SYSTEM (DIRECTIONAL PROCEDURE)

C29.3 DESIGN WIND LOADS: SOLID FREESTANDING WALLS AND SOLID SIGNS

C29.3.1 Solid Freestanding Walls and Solid Freestanding Signs. See commentary, Section C26.10.1.

The Risk Category for rooftop equipment or appurtenances is required to be not less than that for the building on which the equipment is located, nor that for any other facility to which the equipment provides a necessary service. For example, if a solar array is located on the roof of a hospital, design wind load for the solar array is based on the Risk Category for the hospital, even if the solar array is not needed for functioning of the hospital. In another example, if an antenna provides critical communication service for a hospital and the antenna is located on top of a parking structure separate from the hospital building, design wind load for the antenna is based on the Risk Category for the hospital, which is greater than the Risk Category for the parking structure.

Fig. 29.3-1. The provisions in Fig. 29.3-1 are based on the results of boundary-layer wind tunnel studies (Letchford 1985, 2001; Holmes 1986; Letchford and Holmes 1994; Ginger et al. 1998a, b; Letchford and Robertson 1999; Mehta et al. 2012).

A surface curve fit to Letchford's (2001) and Holmes's (1986) area averaged mean net pressure coefficient data (equivalent to mean force coefficients in this case) is given by the following equation (Fox and Levitan 2005):

$$C_f = \{1.563 + 0.008542 \ln(x) - 0.06148y + 0.009011 [\ln(x)]^2 - 0.2603y^2 - 0.08393y[\ln(x)]\} / 0.85$$

where $x = B/s$ and $y = s/h$.

The 0.85 term in the denominator modifies the wind tunnel-derived force coefficients into a format where the gust-effect factor as defined in Section 26.11 can be used.

Force coefficients for Cases A and B were generated from the preceding equation, then rounded off to the nearest 0.05. That equation is only valid within the range of B/s and s/h ratios given in the figure for Cases A and B.

Of all the pertinent studies on single-faced signs, only Letchford (2001) specifically addressed eccentricity (i.e., Case B). Letchford reported that his data provided a reasonable match to Cook's (1990) recommendation for using an eccentricity of 0.25 times the average width of the sign. However, the data were too limited in scope to justify changing the existing eccentricity value of 0.2 times the average width of the sign, which is also used in the 2011 Australian/New Zealand Standard (Standards Australia 2011).

Mehta et al. (2012) tested a variety of aspect ratios (B/s) and clearance ratios (s/h) for double-faced signs with all sides

enclosed to address current industry practice. The study included both wind tunnel testing and a full-scale field test to calibrate the wind tunnel models (Zuo et al. 2014; Smith et al. 2014). These sign configurations exhibited an average reduction of 16% in mean force coefficients with a range of 9% to 22% as compared to single-faced sign force coefficients given by the equation above. These tests also showed that the eccentricity of 0.2 times the width of the structure is overly conservative. Eccentricities reported in the study ranged from 0.039 to 0.105 times the width of the structure, with an average of 0.061. Testing by Giannoulis et al. (2012) supports the findings in Mehta et al. (2012).

Case C was added to account for the higher pressures observed in both wind tunnel studies (Letchford 1985, 2001; Holmes 1986; Letchford and Holmes 1994; Ginger et al. 1998a, b; Letchford and Robertson 1999) and full-scale studies (Robertson et al. 1997) near the windward edge of a freestanding wall or sign for oblique wind directions. Linear regression equations were fit to the local mean net pressure coefficient data (for wind direction 45°) from the referenced wind tunnel studies to generate force coefficients for square regions starting at the windward edge. Pressures near this edge increase significantly as the length of the structure increases. No data were available on the spatial distribution of pressures for structures with low aspect ratios ($B/s < 2$).

The sample illustration for Case C at the top of Fig. 29.3-1 is for a sign with an aspect ratio $B/s = 4$. For signs of differing B/s ratios, the number of regions is equal to the number of force coefficient entries located below each B/s column heading.

For oblique wind directions (Case C), increased force coefficients have been observed on above ground signs compared to the same aspect ratio walls on ground (Letchford 1985, 2001; Ginger et al. 1998a). The ratio of force coefficients between aboveground and on-ground signs (i.e., $s/h = 0.8$ and 1.0, respectively) is 1.25, which is the same ratio used in the Australian/New Zealand Standard (Standards Australia 2002). Note 5 of Fig. 29.3-1 provides for linear interpolation between these two cases.

For walls and signs on the ground ($s/h = 1$), the mean vertical center of pressure ranged from $0.5h$ to $0.6h$ (Holmes 1986; Letchford 1989; Letchford and Holmes 1994; Robertson et al. 1995, 1996; Ginger et al. 1998a); $0.55h$ was the average value. For aboveground walls and signs, the geometric center best represents the expected vertical center of pressure.

The reduction in C_f caused by porosity (Note 2) follows a recommendation (Letchford 2001). Both wind tunnel and full-scale data have shown that return corners significantly reduce the net pressures in the region near the windward edge of the wall or sign (Letchford and Robertson 1999).

C29.3.2 Solid Attached Signs. Signs attached to walls and subject to the geometric limitations of Section 29.3.2 should experience wind pressures approximately equal to the external pressures on the wall to which they are attached. The dimension requirements for signs supported by frameworks, where there is a small gap between the sign and the wall, are based on the collective judgment of the committee.

C29.4 DESIGN WIND LOADS: OTHER STRUCTURES

Guidance for determining G , C_f , and A_f for structures found in petrochemical and other industrial facilities that are not otherwise addressed in ASCE 7 can be found in *Wind Loads for Petrochemical and Other Industrial Facilities* (ASCE Task Committee on Wind-Induced Forces (2011)).

Figs. 29.4-1, 29.4-2, and 29.4-3. With the exception of Fig. 29.4-3, the pressure and force coefficient values in these tables are unchanged from ANSI A58.1-1972 (ANSI 1972). The coefficients specified in these tables are based on wind tunnel tests conducted under conditions of uniform flow and low turbulence, and their validity in turbulent boundary-layer flows has yet to be completely established. Additional pressure coefficients for conditions not specified herein may be found in two references (SIA 1956; ASCE 1961).

With regard to Fig. 29.4-3, the force coefficients are a refinement of the coefficients specified in ANSI A58.1-1982 (1982) and in ASCE 7-93 (1994). The force coefficients specified are offered as a simplified procedure that may be used for trussed towers and are consistent with force coefficients given in TIA (1991) and force coefficients recommended by Working Group No. 4 (IASS 1981).

It is not the intent of this standard to exclude the use of other recognized literature for the design of special structures, such as transmission and telecommunications towers. Recommendations for wind loads on tower guys are not provided as in previous editions of the standard. Recognized literature should be referenced for the design of these special structures as is noted in Section 29.1.3. For the design of flagpoles, see NAAMM (2007). For the design of structural supports for highway signs, luminaires, and traffic signals, see AASHTO LTS-6 (AASHTO 2013).

C29.4.1 Rooftop Structures and Equipment for Buildings. Wind loads on rooftop structures and equipment are revised in ASCE 7-16 to use Eqs. (29.4-2) and (29.4-3) for buildings of all heights. The change provides an improved representation of the limited amount of research that is available (Hosoya et al. 2001; Kopp and Traczuk 2007). The change also eliminates inconsistencies between equipment on roofs below versus slightly above the 60-ft (18.3 m) height. The research in Hosoya et al. (2001) only treated one value of A_f ($0.04Bh$). The research in Kopp and Traczuk (2008) treated values of $A_f = 0.02Bh$ and $0.03Bh$, and values of $A_r = 0.0067BL$. Because GC_r is expected to approach 1.0 as A_f or A_r approaches that of the building (Bh or BL), a linear interpolation is included as a way to avoid a step function in load if the designer wants to treat other sizes. However, the loads provided by these provisions are best suited for units that are much smaller than the building, $A_f < 0.05Bh$ and $A_r < 0.01BL$. The resulting loads are expected to be overly conservative if applied to linelike structures that extend more than $0.1B$ or $0.1L$ across the roof.

Both research studies showed high uplift forces on the rooftop equipment. Hence, uplift loads are addressed in Section 29.4.1.

Mechanical equipment screens commonly are used to conceal plumbing, electrical, or mechanical equipment from view

and are defined as rooftop structures not covered by a roof and located away from the edge of the building roof such that they are not considered parapets. Many configurations and types of screens are available ranging from solid walls to porous panels, which allow some air to flow through. Though the use of equipment screens is prevalent, little research is available to provide guidance for determining wind loads on screen walls and equipment behind screens. Accordingly, rooftop screens, equipment behind screens, and their supports and attachments to buildings should be designed for the full wind load determined in accordance with Section 29.4.1. Where substantiating data have been obtained using the Wind Tunnel Procedure (Chapter 31), design professionals may consider wind load reductions in the design of rooftop screens and equipment. For example, studies by Zuo et al. (2011) and Erwin et al. (2011) suggest that wind loads on some types of screen materials and equipment behind screens may be overestimated by the equations defined in Section 29.4.

The design wind forces for ground-mounted tanks or similar structures (smooth surface such as concrete or steel) with the aspect ratios H/D (height to diameter) in the range of 0.25 to 4 inclusive shall be determined in accordance with Section 29.4.2.

C29.4.2 Design Wind Loads: Circular Bins, Silos, and Tanks with $h \leq 120$ ft ($h \leq 36.5$ m), $D \leq 120$ ft ($D \leq 36.5$ m), and $0.25 \leq H/D \leq 4$. Section 29.4 contains the provisions for determining wind loads on silo and tank walls and roofs. The provisions are largely based on Standards Australia (2011) and the wind tunnel tests of low-rise cylindrical structures carried out at high Reynolds numbers ($Re > 1.0 \times 10^5$) by Sabransky and Melbourne (1987) and Macdonald et al. (1988, 1990). Significant increases in drag forces of grouped silos were found in the wind tunnel tests, so the provisions of grouped tanks and silos are specified in this section.

C29.4.2.1 External Walls of Isolated Circular Bins, Silos, and Tanks. This section specifies the drag coefficient, C_f , for the walls of circular bins, silos, and tanks. The drag coefficient is adopted from Standards Australia (2011). Note that the drag force of 0.63 obtained from an integration of the equations in AS/NZS 1170.2 (Standards Australia 2002) is close to the value of smooth surface type from Fig. 29.4-1.

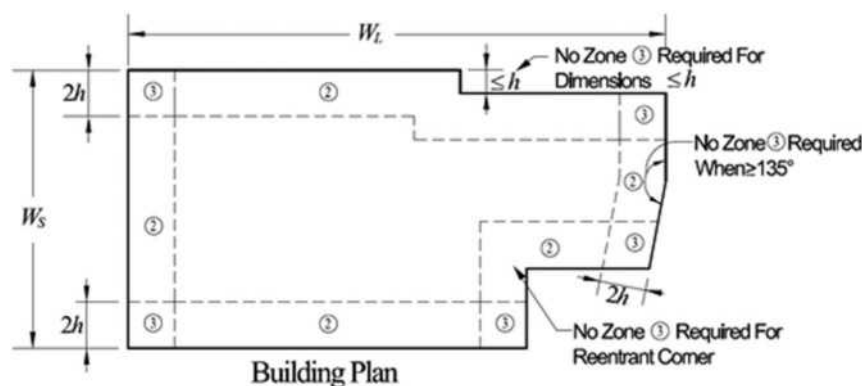
C29.4.2.2 Roofs of Isolated Circular Bins, Silos, and Tanks. This section specifies the external pressure coefficients (C_p) for the roofs of circular bins, silos, and tanks. Two conditions are covered as shown in Fig. 29.4-5: roofs that have the roof angle $\alpha < 10^\circ$, and roofs that have $10^\circ \leq \alpha < 30^\circ$. The results of $\alpha < 10^\circ$ roofs are based on comparisons of domed roofs and flat roofs from Chapter 27 of ASCE 7-10 for maximum uplift conditions. The results of $10^\circ \leq \alpha < 30^\circ$ roofs are consistent with data of Sabransky and Melbourne (1987) and Macdonald et al. (1988). According to the wind tunnel tests, only suctions are observed for the roofs defined.

C29.4.2.3 Undersides of Isolated Elevated Circular Bins, Silos, and Tanks. This section specifies the external pressure coefficients, C_p , for the underneath sides of circular bins, silos, and tanks. The external pressure coefficients, C_p , are adopted from Standards Australia (2011).

For calculating gust-effect factor G , structural period T should be based on the analysis of the whole structure: tank and support structure.

C29.4.2.4 Roofs and Walls of Grouped Circular Bins, Silos, and Tanks. For grouped silos, C_p and C_f values for roofs and walls are largely based on Standards Australia (2011) and wind

Diagram



Note: h = height of building, in ft (m); W_L = width of a building on its longest side, in ft (m); W_S = width of a building on its shortest side, in ft (m)

FIGURE C29.4-1 Sketch of Building with Irregular Plan Dimensions for Use with Wind Loads on Roof-Mounted Solar Panels

tunnel tests by Sabransky and Melbourne (1987) and Macdonald et al. (1990). Test results of an in-line group of three silos with a clear spacing of $0.25D$ between nearest adjacent walls ($1.25D$ center-to-center) by Sabransky and Melbourne (1987) indicated that the drag coefficient for the central cylinder in the group increased by approximately 65% relative to that of the isolated cylinder. Pressure coefficients over the entire roof were significantly larger in magnitude than the corresponding pressure coefficients on the isolated model. It was concluded that a clear spacing of $0.25D$ produced the maximum interference between two finite cylinders.

C29.4.3 Rooftop Solar Panels for Buildings of All Heights with Flat Roofs or Gable or Hip Roofs with Slopes Less Than 7° . Section 29.4.3 addresses low-profile solar panels on low-slope roofs. Requirements specific to this type of solar array are provided because such systems (a) are in widespread use and (b) have been subject to wind tunnel testing that provides guidance on appropriate design wind loads. Fig. 29.4-7 intentionally has a limited range of application, with the maximum height above the roof surface (h_2) for the solar panels limited to 4 ft (1.2 m) and the panel chord length (L_p) limited to 6.7 ft (2.0 m). Wind tunnel data (e.g., Kopp 2013) show that increasing the overall height above the roof or panel chord length increases the wind loads. Likewise, the height of the gap between the panels and the roof surface (h_1) is limited to 2 ft (0.61 m); otherwise, the wind flow under the panels can cause uplift exceeding that covered in the figure. The requirements are not applicable to open structures because the applicable test data are from enclosed structures, which have different aerodynamics than open structures. The roof zones shown in Fig. 29.4-7 are larger than those for the roof loads, as explained in Banks (2012) and Kopp (2013). Buildings with nonrectangular plans, such as that sketched in Fig. C29.4-1, adapted from SEAOC (2012), can be used for guidance in applying the requirements. Reentrant corners do not create a Zone 3 because cornering vortices only form at outward or protruded corners. Similarly, outward corners at angles greater than 90° tend to weaken the vortices; as this angle becomes more obtuse, the building corner begins to flatten out and resemble more of an edge condition.

The nominal net pressure coefficient (GC_{rn})_{nom} curves are derived from wind tunnel test data within the range of parameters allowed by Fig. 29.4-7. These curves are created based on a methodology consistent with that used for the ASCE 7 components and cladding (C&C) loads. The net pressure coefficients shown in the design curves of the figure are denoted (GC_{rn})_{nom}

since these values need to be adjusted for array edge conditions, parapet size, and solar panel length.

The wind tunnel data indicate that the (GC_{rn})_{nom} values are not linearly related to the panel tilt angle over the full tilt angle range. The data indicate that there is a relatively small change in (GC_{rn})_{nom} values for the lower tilt panels in the 1° to 5° range. Then there is a rapid increase in (GC_{rn})_{nom} values from 5° to 15° . There is again a relatively small change in (GC_{rn})_{nom} values for higher tilt panels in the 15° to 35° range because, for the higher tilt angles, upstream panels create turbulence, which increases the wind loads on all downstream panels (e.g., Kopp et al. 2012; Kopp 2013). Thus, the figure was created with two (GC_{rn})_{nom} curves to address this phenomenon: a (GC_{rn})_{nom} curve for low-tilt panels in the 0° to 5° range and another for high-tilt panels in the 15° to 35° range. For panel tilt angles in the 5° to 15° range, linear interpolation is permitted.

(GC_{rn})_{nom} values are for both positive and negative values. Wind tunnel test data show similar positive and negative pressures for solar panels (which are very different than typical roof member design wind loads).

Parapets typically worsen the wind loads on solar panels, particularly on wider buildings. The parapets lift the vortices higher above the roof surface and push them closer together, inward from the edges. It is not entirely clear why the vortex effects are more severe in this situation, but tests show that this can result in wind loads that are significantly greater than in the absence of a parapet, particularly for unshrouded tilted panels. The parapet height factor, γ_p , accounts for this effect.

Solar panels are typically installed in large arrays with closely spaced rows, and the end rows and panels experience larger wind pressures than interior panels, which are sheltered by adjacent panels. To account for the higher loading at the end rows and panels, an array edge increase factor is applied, taken from SEAOC (2012). However, single rows of solar panels can be determined using this section, taking into account that all solar panels are defined as being exposed.

Rooftop equipment and structures, such as HVAC units, screens, or penthouses, can provide some sheltering benefits to solar arrays located directly downwind of the object; conversely, however, the regions around edges of such structures can have accelerated wind flow under varied wind directions. Accordingly, the edge increase factor ignores such structures and is calculated based on the distance to the building edge or adjacent array, neglecting any intervening rooftop structures. This results in the panels adjacent to rooftop objects being designed for higher wind loads to account for the accelerated wind flow.

The requirements can be used for arrays in any plan orientation relative to building axes or edges; the dimensions d_1 and d_2 are measured parallel to the principal axes of the array being considered. The requirement in Fig. 29.4-7 for array panels to be set back from the roof edge is meant to ensure that the panels are out of the high-speed wind in the separated shear layers at the edge. If the array is made up of a single row of solar panels, or a single panel, then d_2 is undefined and $\gamma_E = 1.5$.

Wind tunnel studies have shown that the wind loads on rooftop solar panels need not be applied simultaneously to the roof C&C wind loads for portions of the roof that are covered by the panel. Where a portion of the span of a roof member is covered by a solar array and the remainder is not covered, then the roof member should be designed with the solar array wind load on the covered portion with simultaneous application of roof C&C load on the uncovered portion. In a separate load case, the member should also be checked for C&C wind loads assuming that the photovoltaic panels are not present. For installations of new panels on existing buildings, this separate load case to check the capacity of the existing roof structure to resist the roof C&C wind loads applied over the entire roof area (i.e., assuming that the solar panels are not present) is not required.

The wind loads here were obtained for solar arrays without aerodynamic treatments such as shrouds or deflectors. Uplifting wind load for arrays with shrouding may be lower, but because of the range of possible results and sensitivity to design details, such arrays would need to be wind tunnel tested in order to use reduced loads from those specified here. It should also be noted that horizontal (drag) loads could increase with the use of shrouds or deflectors.

Procedure for Using Fig. 29.4-7. To simplify the use of the figure, the following is a step-by-step procedure.

- Step 1: Confirm applicability of the figure to the solar installation and building.
- Step 2: For panels with $\omega \leq 2^\circ$ and $h_2 \leq 10$ in. (254 mm), the procedure using d_2 Section 29.4.4 per Note 4 may be used.
- Step 3: Confirm that layout provides minimum distance from roof edge per Note 5.
- Step 4: Determine roof zones.
- Step 5: Determine effective wind area and normalized wind area for each element being evaluated.
- Step 6: Compute $(GC_{rn})_{nom}$ from applicable chart, using linear interpolation for values of ω between 5° and 15° .
- Step 7: Apply chord length adjustment factor, γ_c .
- Step 8: Apply the Edge Factor d_2 , γ_E , if necessary.
- Step 9: Apply parapet height factor, γ_p .
- Step 10: Calculate (GC_{rn}) .
- Step 11: Calculate pressure, p , using Eq. (29.4-5).

C29.4.4 Rooftop Solar Panels Parallel to the Roof Surface on Buildings of All Heights and Roof Slopes. Wind loads of roof-mounted, planar solar panels that are close to and parallel to the roof surface tend to be lower than the loads on a bare roof because of pressure equalization (Kopp et al. 2012; Kopp 2013), except on the perimeter of the array. The solar array pressure equalization factor, γ_a , accounts for this reduction, based, in particular, on data from Stenabaugh et al. (2015). For pressure equalization to occur, the panels cannot be too large, there needs to be a minimum gap between the panels, and the height above the roof surface cannot be too large. The current requirements are based on panel sizes up to 6.7 ft (2.0 m) long for heights above the roof surface that are less than 10 in. (254 mm) and a minimum gap around the panels of 0.25 in. (6.35 mm). Larger gaps and

lower heights above the roof surface could further decrease the wind loads, but wind tunnel testing would be required to take advantage of this difference. For metal roof panels, the 10-in. (254-mm) maximum distance above the roof surface is measured from the flat portion of the panels, rather than from the top of the panel ribs.

Panels around the edge of the array may experience higher wind loads. The definition of these exposed panels is the same as for tilted panels in Section 29.4.3.

C29.5 PARAPETS

Before the 2002 edition of ASCE 7, no provisions for the design of parapets were included because of the lack of direct research. In the 2002 edition of ASCE 7, a rational method was added based on the committee's collective experience, intuition, and judgment. In the 2005 edition, the parapet provisions were updated as a result of research performed at the University of Western Ontario (Mans et al. 2000, 2001) and at Concordia University (Stathopoulos et al. 2002a, b).

Wind pressures on a parapet are a combination of wall and roof pressures, depending on the location of the parapet and the direction of the wind (Fig. C29.5-1). A windward parapet experiences the positive wall pressure on its front surface (exterior side of the building) and the negative roof edge zone pressure on its back surface (roof side). This behavior is based on the concept that the zone of suction caused by the wind stream separation at the roof eave moves up to the top of the parapet when one is present. Thus, the same suction that acts on the roof edge also acts on the back of the parapet.

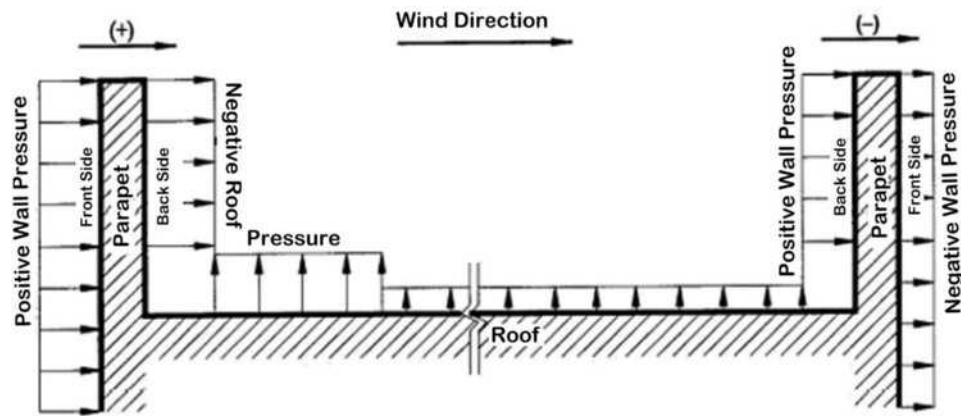
The leeward parapet experiences a positive wall pressure on its back surface (roof side) and a negative wall pressure on its front surface (exterior side of the building). There should be no reduction in the positive wall pressure to the leeward parapet caused by shielding by the windward parapet because, typically, they are too far apart to experience this effect. Because all parapets would be designed for all wind directions, each parapet would in turn be the windward and leeward parapet and, therefore, must be designed for both sets of pressures.

For the design of the main wind force resisting system (MWFRS), the pressures used describe the contribution of the parapet to the overall wind loads on that system. For simplicity, the front and back pressures on the parapet have been combined into one coefficient for MWFRS design. The designer should not typically need the separate front and back pressures for MWFRS design. The internal pressures inside the parapet cancel out in the determination of the combined coefficient. The summation of these external and internal, front and back pressure coefficients is a new term (GC_{pn}) , the combined net pressure coefficient for a parapet.

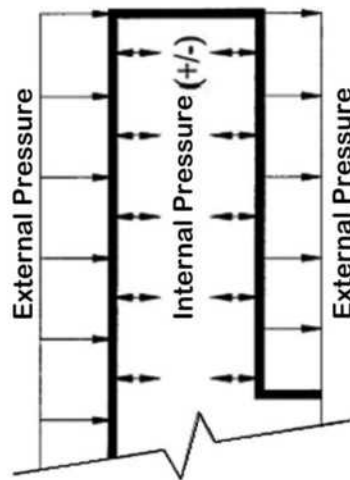
For the design of the components and cladding (C&C), a similar approach was used. However, it is not possible to simplify the coefficients because of the increased complexity of the C&C pressure coefficients. In addition, the front and back pressures are not combined because the designer may be designing separate elements on each face of the parapet. The internal pressure is required to determine the net pressures on the windward and leeward surfaces of the parapet. The provisions guide the designer to the correct (GC_p) and velocity pressure to use for each surface, as illustrated in Fig. C29.5-1.

Interior walls that protrude through the roof, such as party walls and fire walls, should be designed as windward parapets for both MWFRS and C&C.

The internal pressure that may be present inside a parapet is highly dependent on the porosity of the parapet envelope.



Methodology used to Develop External Parapet Pressures
(Main Wind Force Resisting Systems and Components and Cladding)



External and Internal Parapet Pressures
(Component and Cladding Only)

FIGURE C29.5-1. Design Wind Pressures on Parapets

In other words, it depends on the likelihood of the wall surface materials to leak air pressure into the internal cavities of the parapet. For solid parapets, such as concrete or masonry, the internal pressure is zero because there is no internal cavity. Certain wall materials may be impervious to air leakage, and as such have little or no internal pressure or suction, so using the value of (GC_{pi}) for an enclosed building may be appropriate. However, certain materials and systems used to construct parapets containing cavities are more porous, thus justifying the use of the (GC_{pi}) values for partially enclosed buildings or higher. Another factor in the internal pressure determination is whether the parapet cavity connects to the internal space of the building, allowing the building's internal pressure to propagate into the parapet. Selection of the appropriate internal pressure coefficient is left to the judgment of the design professional.

C29.7 MINIMUM DESIGN WIND LOADING

This section specifies a minimum wind load to be applied horizontally on the entire vertical projection of the other structures, as shown in Fig. C27.1-1. This load case is to be applied as a separate load case in addition to the normal load cases specified in other portions of this chapter.

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